

BASICS

Two Knobs, 44 Models

SHAPE

- SHAPE picks the model; TIMBRE and COLOR vary per model
- OSC switches models per step; TBR and COL move the knobs
- Put an LFO or AHD on TIMBRE or COLOR for movement

Lo-Fi Crunch

DEGRADE · REDUX

- DEGRADE lowers the sample rate, REDUX the bit depth
- DEG and RED in a phrase crunch single notes or fills

ANALOG MODELS	
00	CSAW <i>T notch width · C notch depth (phasing)</i>
01	MORPH <i>T triangle → saw → square → pulse · C tone</i>
02	SAW SQUARE <i>T dephase / pulse width · C saw → square</i>
03	SINE TRIANGLE <i>T wavefolder · C sine / triangle balance</i>
04	BUZZ <i>T sine → comb (formant) · C detune</i>
05—06	SQUARE / SAW SUB <i>T width · C sub -1 or -2 octaves</i>
07—08	SQUARE / SAW SYNC <i>T synced pitch · C mix</i>
09—0C	TRIPLE SAW / SQU / TRI / SIN <i>T, C tune oscillators 2 and 3 (snap to intervals)</i>
0D	TRIPLE RING <i>T, C ring-mod partner pitches</i>
0E	SAW SWARM <i>T detune of 7 saws · C high-pass</i>
0F	SAW COMB <i>T comb pitch · C feedback (80 = none)</i>
10	TOY <i>T clock rate · C circuit-bent glitches</i>

Supersaw Lead

0E SAW SWARM

- Raise TIMBRE until the seven saws beat and spread
- COLOR high-passes the lows so it sits above the bass
- ADSR on volume with a long REL, plus a REV send

Sync Sweep

08 SAW SYNC

- AHD or LFO on TIMBRE sweeps the synced harmonics
- COLOR blends in the plain oscillator to soften it

Sub Bass

05 SQUARE SUB

- COLOR sets the sub one or two octaves down
- Low CUTOFF and a short AHD on volume keep it tight

DIGITAL & VOCAL MODELS

11—14

DIGITAL FILTER LP / PK / BP / HP
T cutoff · C saw → square → triangle

15

VOSIM
T, C the two formant pitches

16—17

VOWEL / VOWEL FOF
T a · e · i · o · u · C formant shift

18

HARMONICS
T center harmonic · C spread (12 sines)

19—1B

FM / FEEDBACK / CHAOTIC
T mod amount · C ratio

Talking Lead

16 VOWEL

- A slow LFO on TIMBRE morphs through a, e, i, o, u
- COLOR with a lower note shifts the voice deeper or older

CZ Zap Bass

11 DIGITAL FILTER LP

- Short AHD on TIMBRE sweeps the built-in cutoff down
- No filter needed; COLOR picks the wave, RES stays at 00

Drawbar Organ

18 HARMONICS

- TIMBRE picks which harmonic is loudest
- Low COLOR is pure and hollow, high COLOR is bright and full
- ADSR on volume with full SUS for a held organ chord

PHYSICAL MODELS

1C

PLUCKED
T damping · C pluck position

1D

BOWED
T friction · C bow position (needs a held note)

1E—1F

BLOWN / FLUTED
T air pressure · C instrument shape

20

STRUCK BELL
T damping · C inharmonicity

21

STRUCK DRUM
T damping · C brightness

Plucked Lead

1C PLUCKED

- TIMBRE sets how long the string rings
- Very low notes become knocks and wood percussion
- A DEL send and PSL slides give a guitar-like line

Bells & Mallets

20 STRUCK BELL

- Low COLOR is sweet and tuned, high COLOR is clangy
- TIMBRE shortens the ring for marimba-like hits

Drone Strings

1D BOWED

- Hold notes with an ADSR on volume; end them with OFF
- A slow LFO on COLOR moves the bow for a living tone

808 DRUM MODELS

22

KICK
T decay · C tone; the note tunes the drum

23

CYMBAL
T band-pass cutoff · C metal ↔ noise

24

SNARE
T tone · C snappy noise

Tuned 808 Kick

22 KICK

- Play it on the song's root note; long TIMBRE for a boomy tail
- Raise AMP with LIM SIN for weight

Closed & Open Hats

23 CYMBAL

- Short AHD on volume for closed, longer DEC for open
- KIL 02 on closed hats; COLOR toward noise for a softer hat

WAVETABLE MODELS

25

WAVETABLES
T sweep the table · C pick 1 of 20 tables

26

WAVE MAP
T, C scan a 16 × 16 grid of waves

27

WAV LINE
T move through waves · C interpolation

28

WAV PARAPHONIC
T morph 16 waves · C 4-voice chord (00 = unison)

Chords on One Track

28 WAV PARAPHONIC

- COLOR picks the chord; COL in the phrase changes it per step
- COLOR 00 stacks detuned unison voices for a fat lead

Scanning Pad

26 WAVE MAP

- Slow LFO on TIMBRE, a slower one on COLOR
- Neighboring waves sound alike, so the scan stays smooth

NOISE & TEXTURE MODELS

29

FILTERED NOISE
Note = cutoff · T resonance · C low-pass ↔ high-pass

2A

TWIN PEAKS NOISE
T resonance · C peak spacing

2B

CLOCKED NOISE
Note = rate · T periodicity · C 2—32 levels

2C—2D

GRANULAR CLOUD / PARTICLE
T grain density · C pitch scatter

2E

DIGITAL MOD
T bit rate · C 8-bit data value

Noise Riser

29 FILTERED NOISE

- The note sets the cutoff, so PSL or PBN up sweeps it
- TIMBRE adds a resonant whistle; COLOR toward HP thins it

Ambient Texture

2C GRANULAR CLOUD

- TIMBRE for grain density, COLOR to scatter the pitch
- Long REV send and slow notes for evolving beds